



Inferences for Two Population Means Transcript

Hello everyone, welcome to lesson 1 video 1. In this session, we'll study inferences for two population means. This means testing whether the differences between two groups is statistically significant or not. The main topics in this video include two sample tests, dependent or independent tests and so on.

In our journey to statistics, we have learnt how to test a claim about a single group. Remember in the last videos, we were discussing about comparing one mean or identifying the sample from one population and we were trying to understand whether that is significant or not because we have a claim, let's say, either the average height of people is greater than 5, greater than 6 of a specific place or let's say average weight or average BP is greater than a certain value. So it's all about a same single population.

What if we have groups, we are now trying to compare groups. Let's say if there is a claim that average height of people living in Lucknow is lesser than average height of people living in Kashmir. So this is a claim.

Now we have two different groups, one is a place in Uttar Pradesh and one is a place in Jammu and Kashmir. Now you are talking about two different populations. What if someone says the price of books online and offline, different books online and offline, they are different or they are not same.

What if someone comes in and makes this claim. Now we are also comparing two different populations, right. One book sold on online and one book sold on offline.

So whenever there is a situation and we want to compare two different populations or more specifically the means of two populations, then we do two dependent tests or let's say independent, independent, two sample tests. Okay. So this is what we will be studying in this entire video.

I hope it is more of a fun to understand all of these things and you are able to relate to everything. Okay. So let's say for example, if we are comparing two different means and you would want to see, is there a difference between them? Remember the example that I took, the height of people living in Kashmir is greater than the height of people living in Lucknow, right? So is there a difference between them? So our entire claim is about whether there is the difference or not, right? So in your null hypothesis, you will say there is no difference in your alternate hypothesis.

You will say there is a difference, okay? So these are different types of scenarios that we generally test for. Does a new teaching method lead to a higher test score than the



old one, right? So you have a old teaching method, you have done testing or you have taken sample on that. You have a new teaching method.

You have a sample for that. Now you are comparing the two methods, okay? Does a new drug reduced blood pressure? So you have existing drug, you know what was its effect on the blood pressure and you have got a new drug and you now know what is its effect on the blood pressure. So now you are comparing the two different types of drugs, right? Do engineering students have a higher average salary than the arts students? So you have two different groups of students, one engineering and one arts and you are comparing whether their salaries are different, right? You have group one versus group two.

In this case, you have before and after wherein you have the kind of people on which it was tested might have been the same because you would want to test whether it does the improvement on same specific set of people, right? So it's a before and after. In this case, the population remains the same, but we have two different methodologies. One before the testing what has happened and one after the testing what has happened.

So I'll give you more intuitions to it in the next slides, but at least you understand that when you are comparing two groups, there could be different types of comparisons. One in which you are comparing two different population altogether, group A versus group B, one in which the same population, but you are testing one type of effect, right? One in which the groups are different and you are trying to see if the average values are different or same in those groups, right? So before we can analyse the data, we must answer one crucial steps, are the two samples independent or paired, right? So in these cases, what do I mean by independent versus paired? In those cases, the null hypothesis is basically either there would be only these two decisions, right? If there is a difference and there is no difference. When we say there is no difference in that case, we are actually considering the null hypothesis because null hypothesis is the one wherein you are aware that there is no difference.

And alternate is when you are specifically looking into that suspicion that you have, when you say that the delivery time of Pizzart is lesser than delivery time of Domino's, that's the suspicion that you have, right? And you would want to test that suspicion. So here in this case, this is the null hypothesis because you can also write $\mu_1 - \mu_2$ is equal to 0, which is equivalent to $\mu_1 - \mu_2$ is equal to 0, right? So μ_1 is equal to μ_2 is equivalent to this. Now, in the example that I was just stating, you would state that average of Pizzart is less than average of Domino's.

That would be your alternate hypothesis, while your null hypothesis would be exactly same as this. Average of Pizzart, which would be your μ_1 is equal to μ_2 , which is your Domino's, right? Now, paired or dependent samples. Paired samples are when the sample or the groups remain the same, it's just that the effects are different.



Let's say if I would want to test the effect of teaching on the same group of people, okay, there are two different types of teaching methods, I would want to say which method is effective. But the group remains the same, I mean, the students on which the method is applied remains the same. Similarly, let's say if I would want to test for a particular medical, in medical terms, a particular medicine is very effective or not, okay, I might want to investigate on the same set of people, right? So there are two different medicines, medicine 1 is applied on same set of people, medicine 2 is applied on the same set of people and you will see whether medicine 1 is effective versus medicine 2 is effective.

So all the cases wherein the group remains the same, okay, the entire group on which the analysis is done remains the same or the sample remains the same. In that case, we have paired samples, okay? So what are the examples? Measuring the weight of same group of people before and after the diet. So you have same group of people, you are measuring their weight before the diet and after the diet and then you are trying to see if the diet programme is actually effective or not, right? Then testing the effect of a skin care cream by applying its patient's left arm and the placebo to the right arm, then also that's a dependency test, okay, a dependent test.

Comparing the performance of two tyres by putting one of each on front wheel and of the same set of the cars, let's say for example, you have same set of the cars, but you are trying to judge the performance of two different types of tyres and then you are placing them at different positions of the car and you are trying to test whether the performance of the tyres is equivalent or not, right? So in all of these examples, if you think of in all of these three examples that I have just stated or I have written down here, the group remains the same, right? The performance of two tyres by putting them on front wheels of the same set of the cars if you think of, so what you are doing is, you are testing it on the same set of tyres by putting the tyres at different different positions, you are testing the performance of the tyres at different different positions, but the tyres remain the same. Similarly, when you are testing for the weight of same group of people before and after a ride, you are testing it on the same group of people, right? You are not changing the group. So when the group remains intact, that's a paired test.

Now this is the crucial decision making process. Just if you would want to know whether you would want to use paired test versus non-paired or independent test, paired test is your dependent test when you see or when you hear about the framing like when someone says it's a paired test or it's a dependent test, both of them mean the same thing. Dependent is paired, paired is dependent, right? So how do you decide that whether it's a paired test or it's a independent test, right? First thing is, you see whether the group is same or not, okay? Is each data point in group one naturally matched to the group two? If that is true, the group remains the same, the set of people remains the same, then you are doing a paired test.



If that group doesn't remain the same, then you are doing an independent test, okay? So that's a very simple distinction between them. Then the formulation for doing the test statistics. So in all of these, we generally take t-tests.

So what we do is, if you see here at the top, when your test is independent type, you have two different groups, you are taking average of group one, average of group two, and you are seeing whether there is a difference in those averages of sample statistics. Minus because you are comparing it against the null hypothesis that there is no difference, hence you are subtracting it from the null hypothesis assumption, okay? So this is what your alternate hypothesis is, that there is a difference, minus this is what your null hypothesis is. If your alternate hypothesis, I mean, there is a difference, then this particular difference between x_1 and x_2 , the first term in the numerator would be significantly higher than zero, okay? So if that becomes significantly higher, t-test value would be really high, okay? In that case, p-value would be very much lesser than 0.05, and hence you will say the alternate hypothesis is true, because obviously your numerator is way higher than zero, which means that the null hypothesis is actually out of the picture, right? Similarly, in paired t-test, what happens? Now, in paired t-test, you are actually comparing the differences.

Let's say the same example we discussed before, we have same group of people, and we are trying to see whether their weights have, what has happened to their weights, right? So if d is after, before and after whatever their weights were, you will take difference between their weights, which can be computed as $\bar{d}$, okay? This is the difference of their weights, minus you assume that in your null hypothesis that the effect of the diet plan is not there, so diet plan is not effective, okay? Now, you have $\bar{d}$ minus zero divided by standard error, you will compute this statistics, if the t-statistics is very, very huge or it is very significant, in that case you will find p-value is less than 0.05, hence you will say that the alternate hypothesis is true, otherwise you will say if d -value is almost similar to zero, in that case d -value would be smaller, hence you will say that the diet plan in this case is not very significant or the diet plan does not affect the weight a lot, okay? So, these are two different types of scenarios, one we discussed about independent t-test and one we discussed about paired t-test, right? This is a simple scenario, you have two different groups and you can obviously check and do interactions here and change the group variability. Now, if you see here, group 1 seems to be very different from group 2, if you see in the value of group 1, first observation is 58.8 versus group 2, 53.5 and then 64.3, 69.6. Now, let's say for example if it is a independent test, in that case you would be doing a two sample independent test and here in this case it comes out to be, it is not significant because you see that there is a difference but the difference is not very significant, okay? I mean there are chances that both the groups are same or both the populations are same, that's why that's what it is saying. But there would be, there might be situations wherein there is a lot of variability, I don't think I am able to bring that.



So, let's say for example group 1's value is significantly higher than group 2's value, a lot higher than. In those cases, you will see that both the groups, the statistical value, the statistics value, p-value or t-value would be much much higher or in this case p-value would be slightly lower than 5%, hence you will say that it is significant, not like the case which is represented here, okay? Now, to summarise what we have learnt, we have learnt about what exactly is a paired test, what exactly is an independent test. Paired test or dependent test is the same thing when you have groups same but the effects applied are different.

So, if you are interested in testing the efficacy of two different medicines on the same group of people, you are doing a paired test. When you are comparing two different groups, let's say when you are comparing people living in India versus people living in US, then that is an independent test because the two groups are entirely different, okay? I hope all of these make sense now and we will get more deeper into numericals and understand all of these things much more in the next video. In summary, testing two population means helps us to compare group in a statistically significant way.

By accounting for variability within each group, we can judge whether the observed difference is meaningful or not. In the next video, we will extend our approach to comparing variances of two populations.

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